

Phenomenological estimate for the longitudinal structure function in SIDIS

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3D Structure of the proton

What is the proton's non-perturbative transverse structure?

→ Encoded by Transverse Momentum Distributions (TMDs)

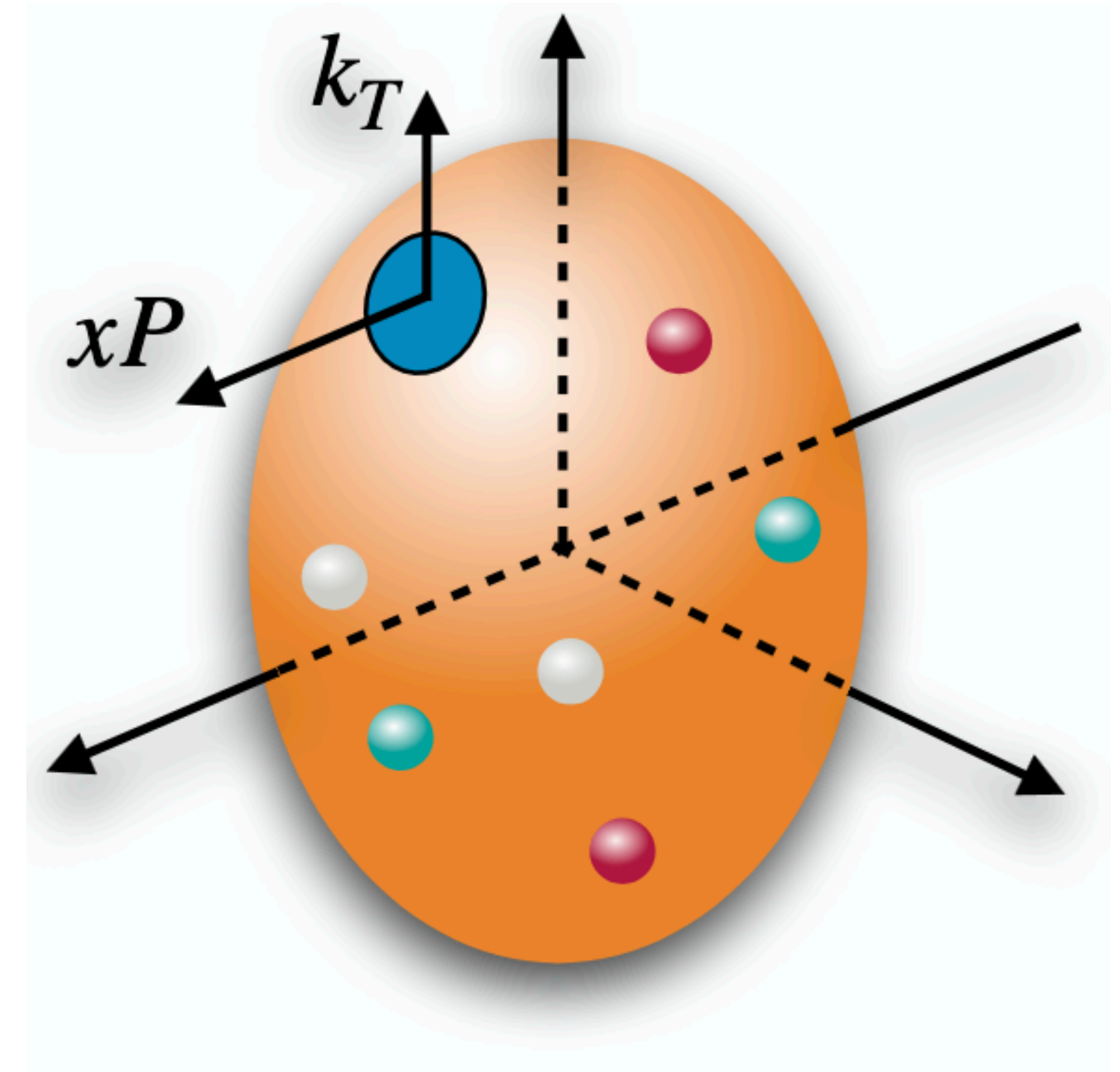
$$f_{i/P}(x, k_T; \mu^2, \zeta)$$

→ collinear backbone:

$$\text{OPE: } f_{i/P}(x, b_T) = [C \otimes f_{i/P}(x)] E(\mu, \zeta) f_{\text{NP}}$$

$$0^{\text{th}} \text{ moment: } 2\pi \int_0^\mu dk_T k_T f_{i/P}(x, k_T) \sim f_{i/P}(x; \mu)$$

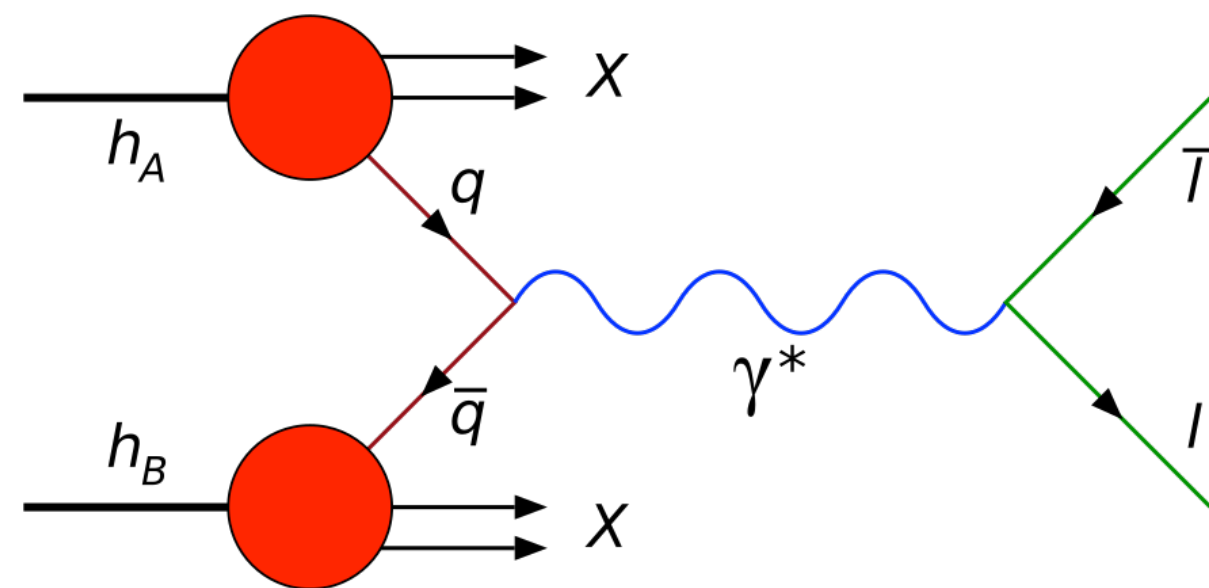
→ transverse space contains much *richer* (and more complicated) dynamics



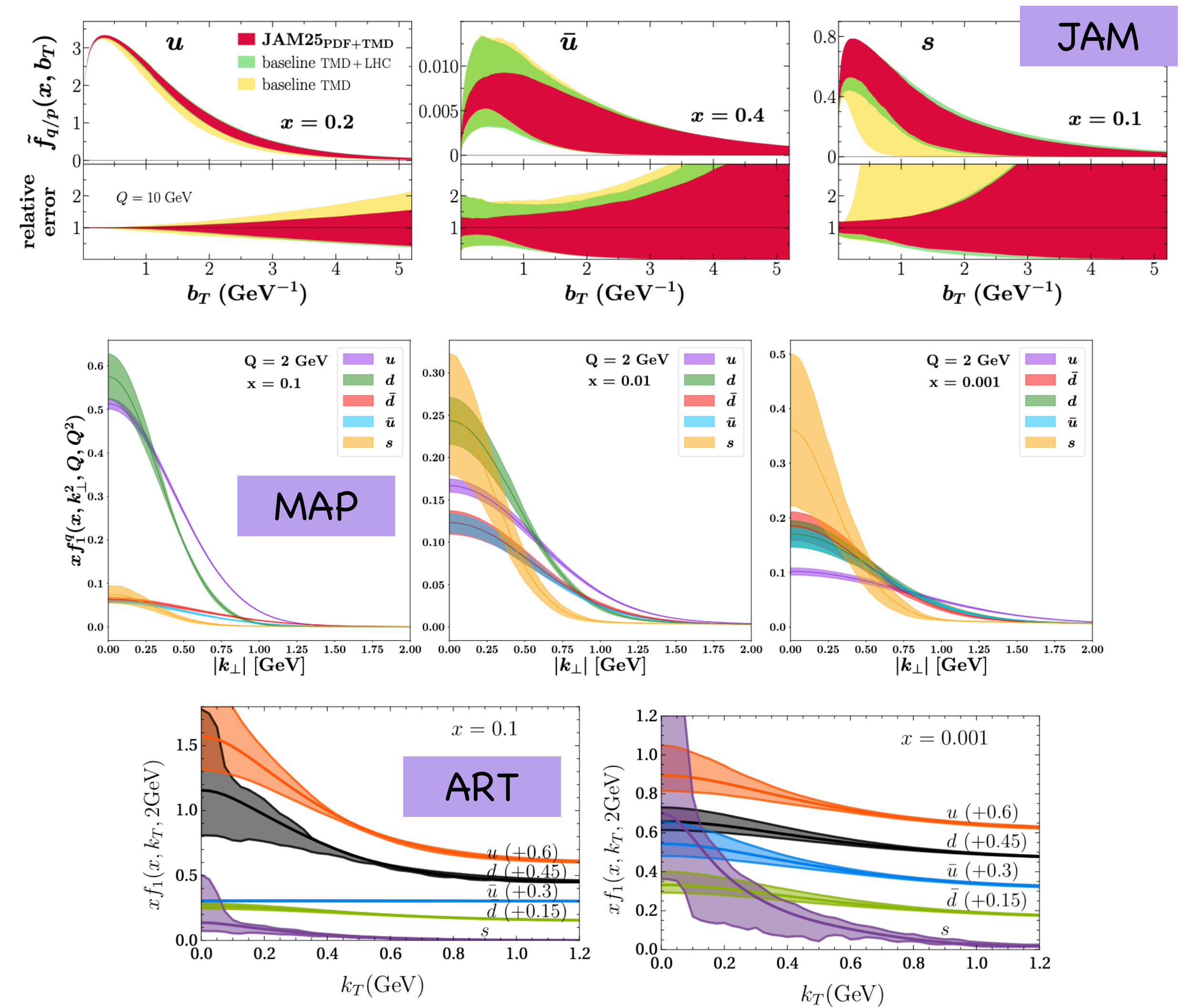
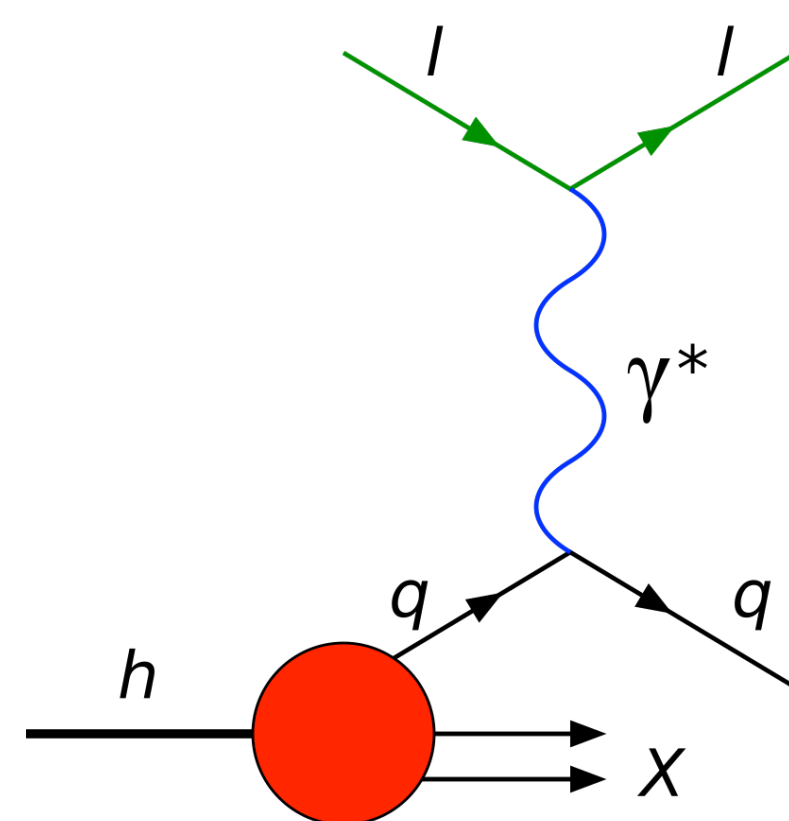
How do we access TMDs?

- 1) Universality* and factorization
- 2) Global Analysis

Drell-Yan



SIDIS



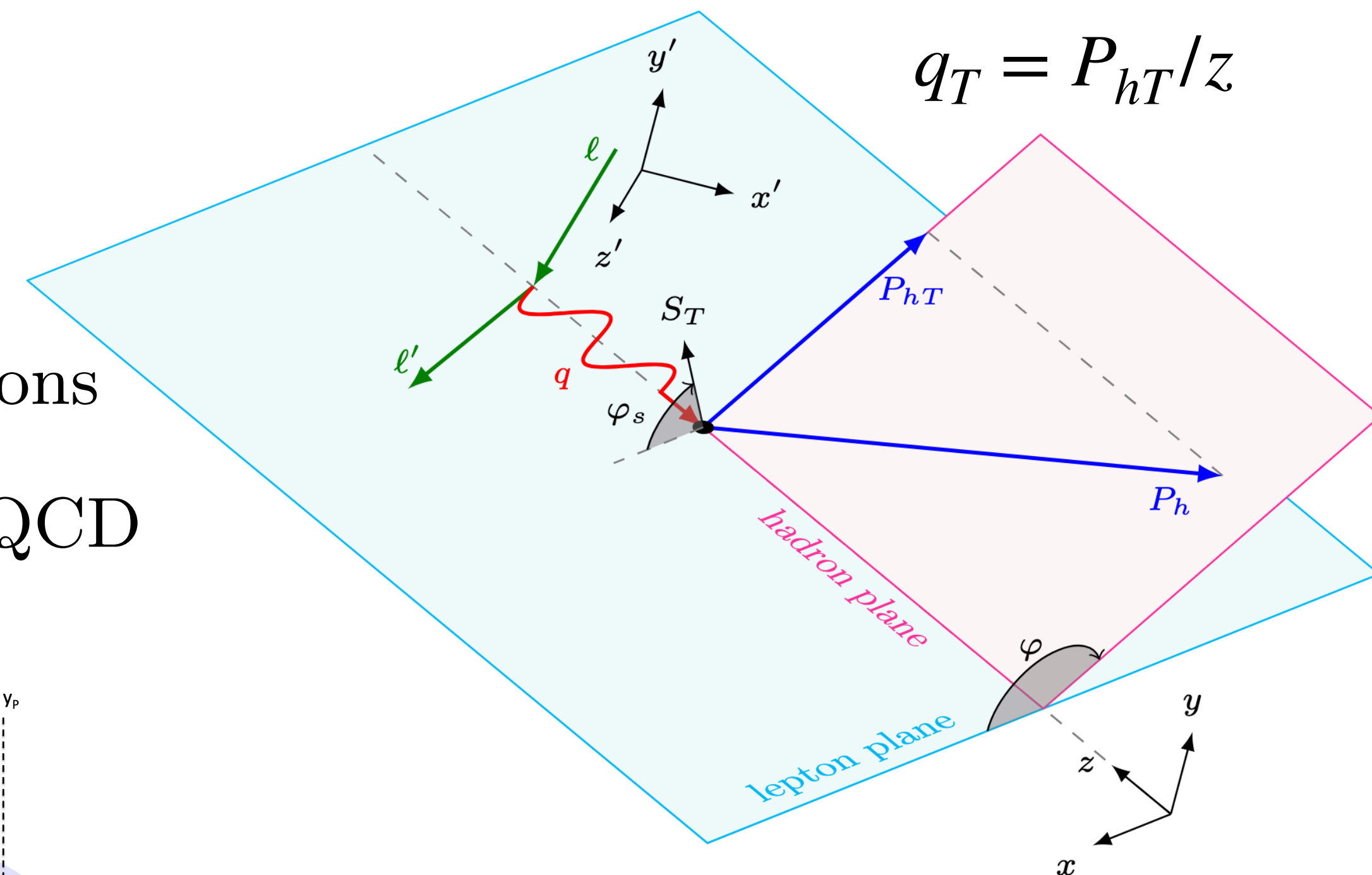
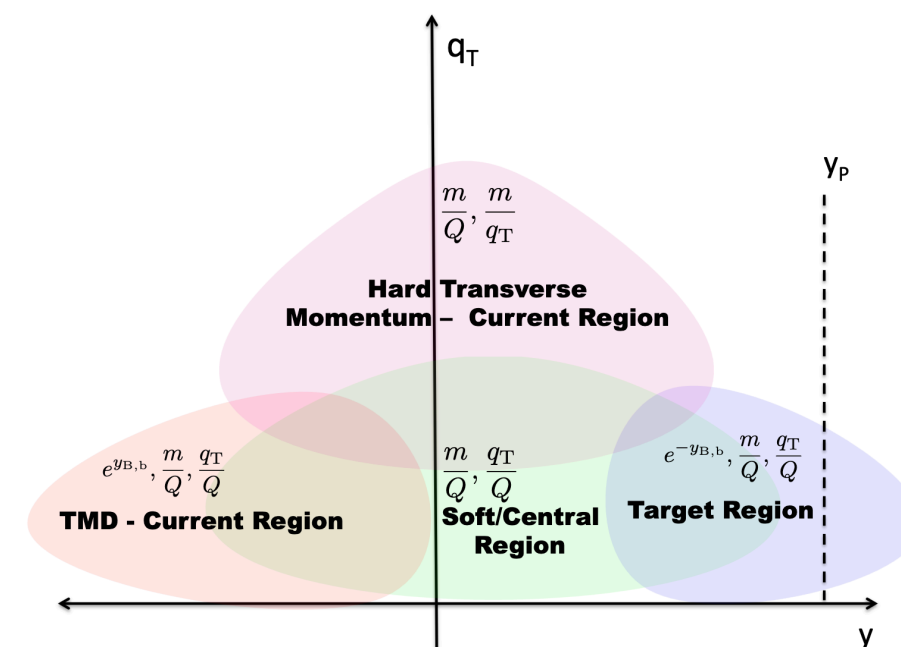
SIDIS basics

$$\frac{d\sigma}{dx dy dz dq_T^2 d\varphi} = \frac{\alpha^2 y}{Q^2(1-\varepsilon)} \left[F_{UU,T} + \varepsilon F_{UU,L} - \sqrt{2\varepsilon(1+\varepsilon)} F_{UU}^{\cos\varphi} \cos\varphi + \varepsilon F_{UU}^{\cos 2\varphi} \cos 2\varphi \right]$$

Observation of final-state hadron*

- Enhanced flavor separation
- Unique correlations encoded in azimuthal modulations
- Bridge between hadron structure, hadronization, pQCD

*produced via current fragmentation



CSS formalism

Scale separation: $F = W + \text{FO} - \text{ASY}$

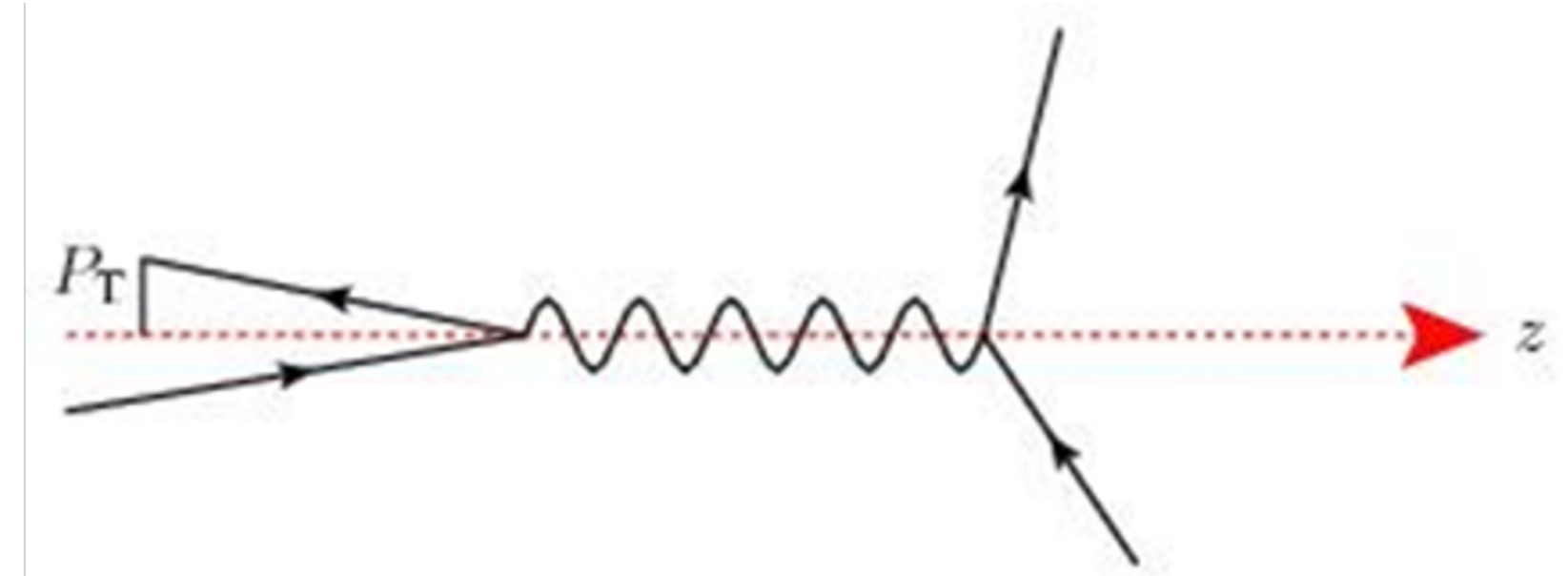


$$F^{(W)} = C[\omega_{T/L} f_1 D_1]$$

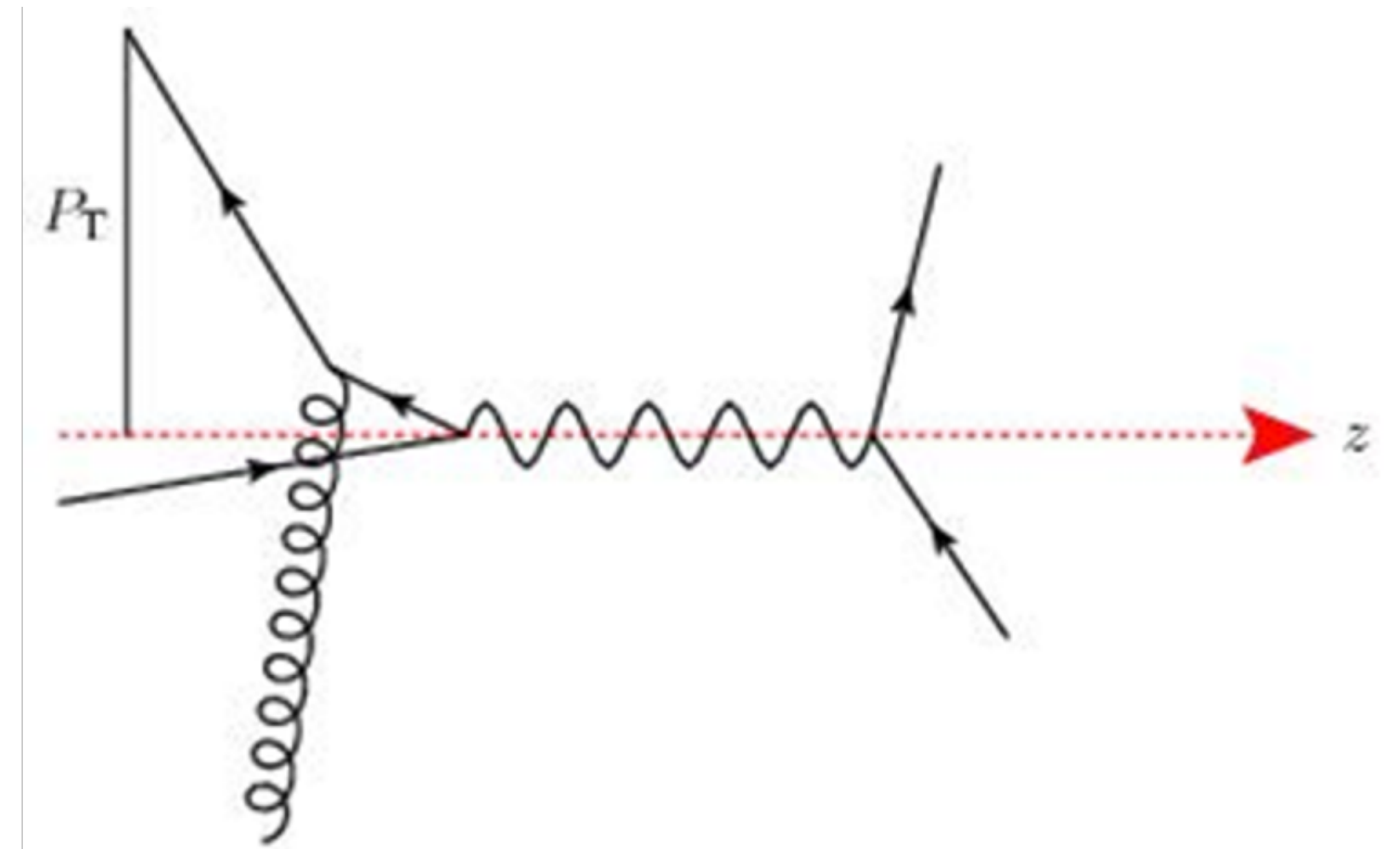
$$F^{(\text{ASY})} = \lim_{q_T/Q \rightarrow \infty} F_{UU,T/L}^{(W)} \approx \lim_{q_T/Q \rightarrow 0} F_{UU,T/L}^{(\text{FO})}$$

$$F^{(\text{FO})} = \frac{1}{Q^2} \frac{\alpha_s}{(2\pi z)^2} \sum_{ij} C_{UU,T/L}^{(ij)} \otimes f_{i/P} \otimes D_{h/j}$$

W term probes *intrinsic* transverse motion



FO captures *hard* radiation

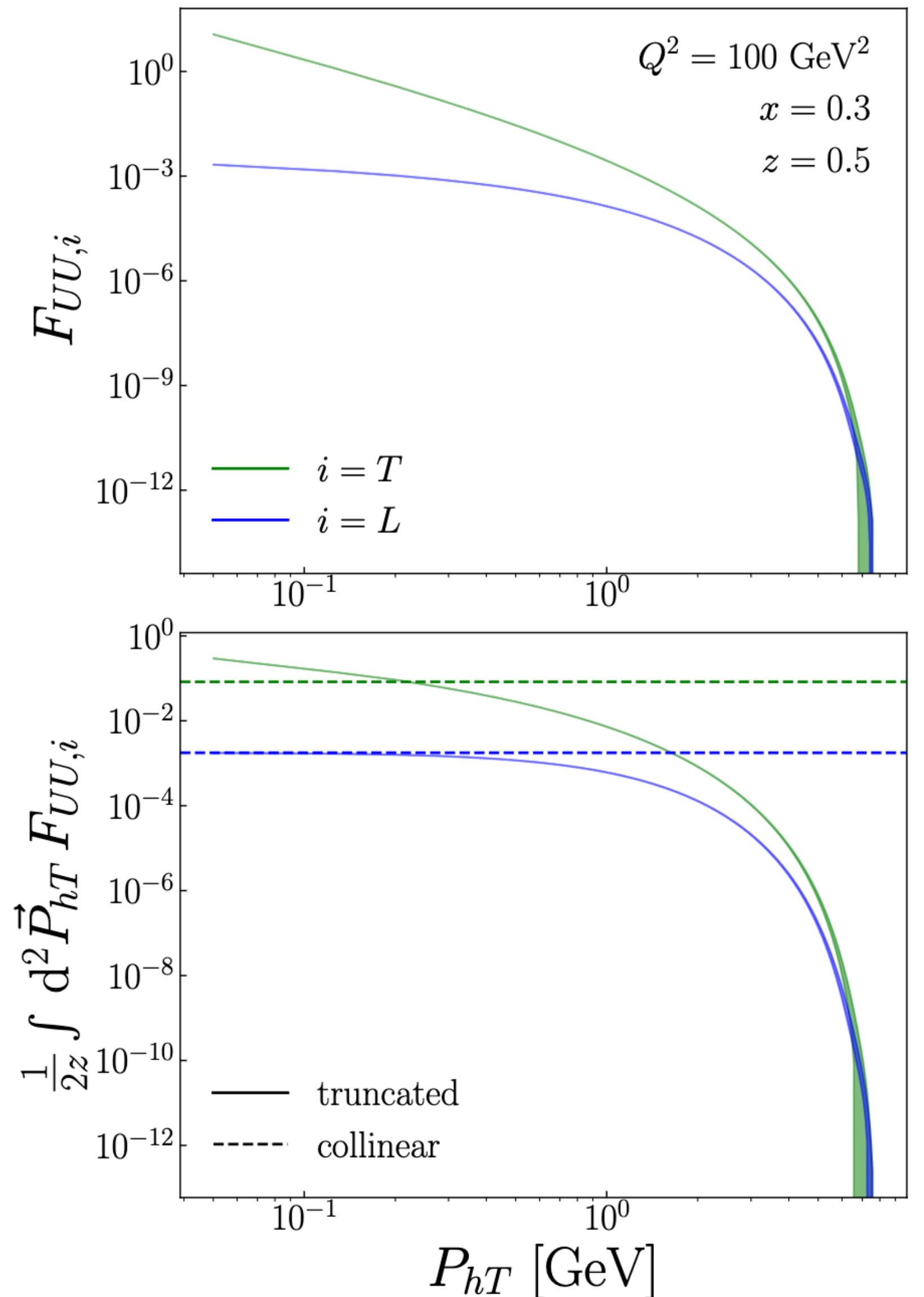


Low q_T behavior and integrability

$$\frac{d\sigma}{dx dQ^2 dz} = \frac{\alpha^2 xy}{Q^2(1-\varepsilon)} [F_T + \varepsilon F_L] = \int d^2 \vec{P}_{hT} \frac{d\sigma}{dx dQ^2 dz d^2 \vec{P}_{hT}}$$

$$\Rightarrow F_{T/L} = \frac{1}{2z} \int d^2 \vec{P}_{hT} F_{UU,T/L}$$

Integral is over all $\vec{P}_{hT}$ with $W+Y$!



Low q_T behavior and integrability

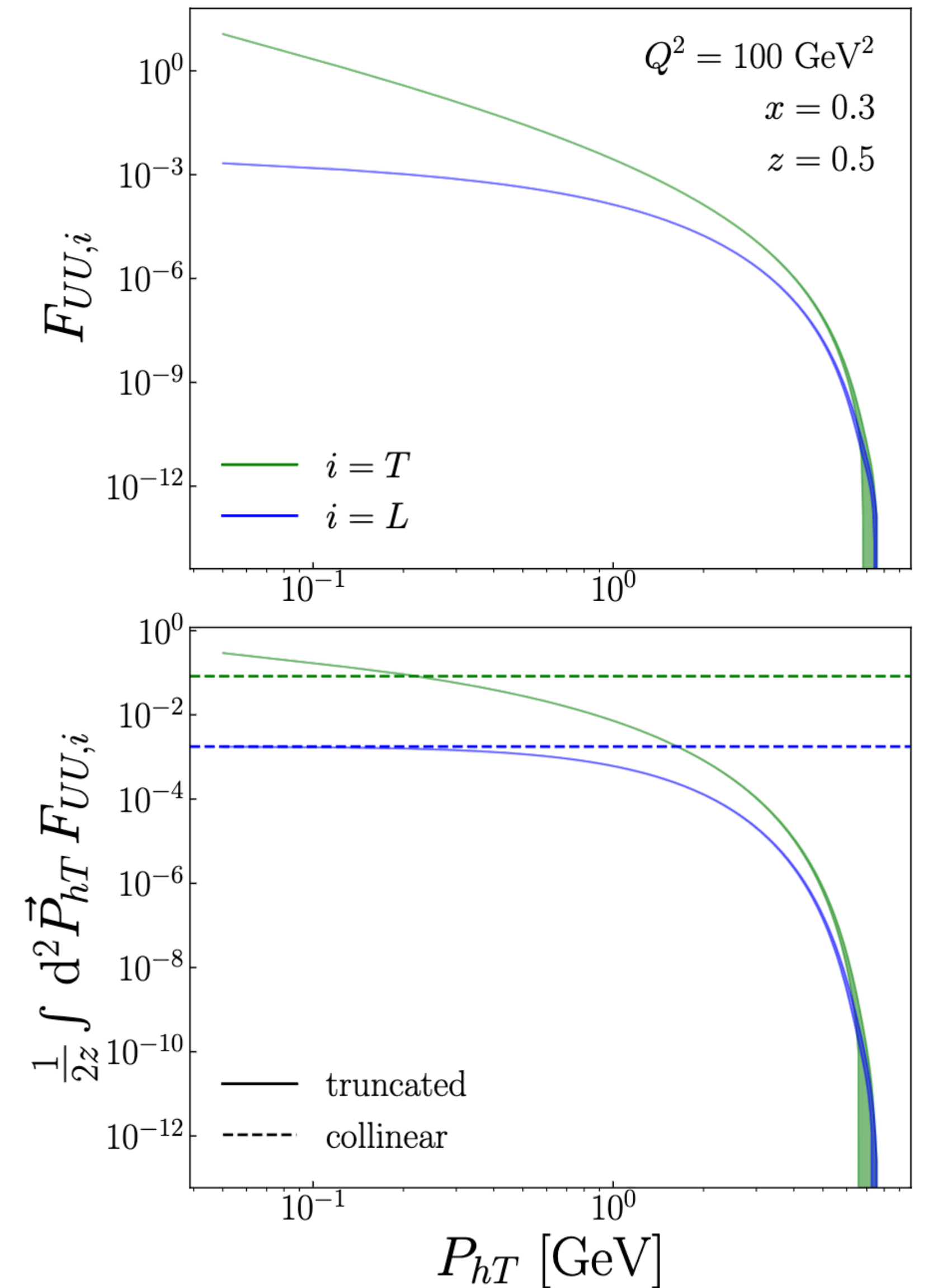
$$F_{UU,T}^{(W)} = C[f_1 D_1] \quad F_{UU,L}^{(W)} = \mathcal{O}\left(\frac{M^2}{Q^2}, \frac{q_T^2}{Q^2}\right)$$

↪ Leading power
↪ Sub-leading power

$$F_{UU,T}^{(FO)} \sim \frac{1}{q_T^2} \ln(Q^2/q_T^2) \quad F_{UU,L}^{(FO)} \sim \frac{1}{Q^2} \ln(Q^2/q_T^2)$$

↓

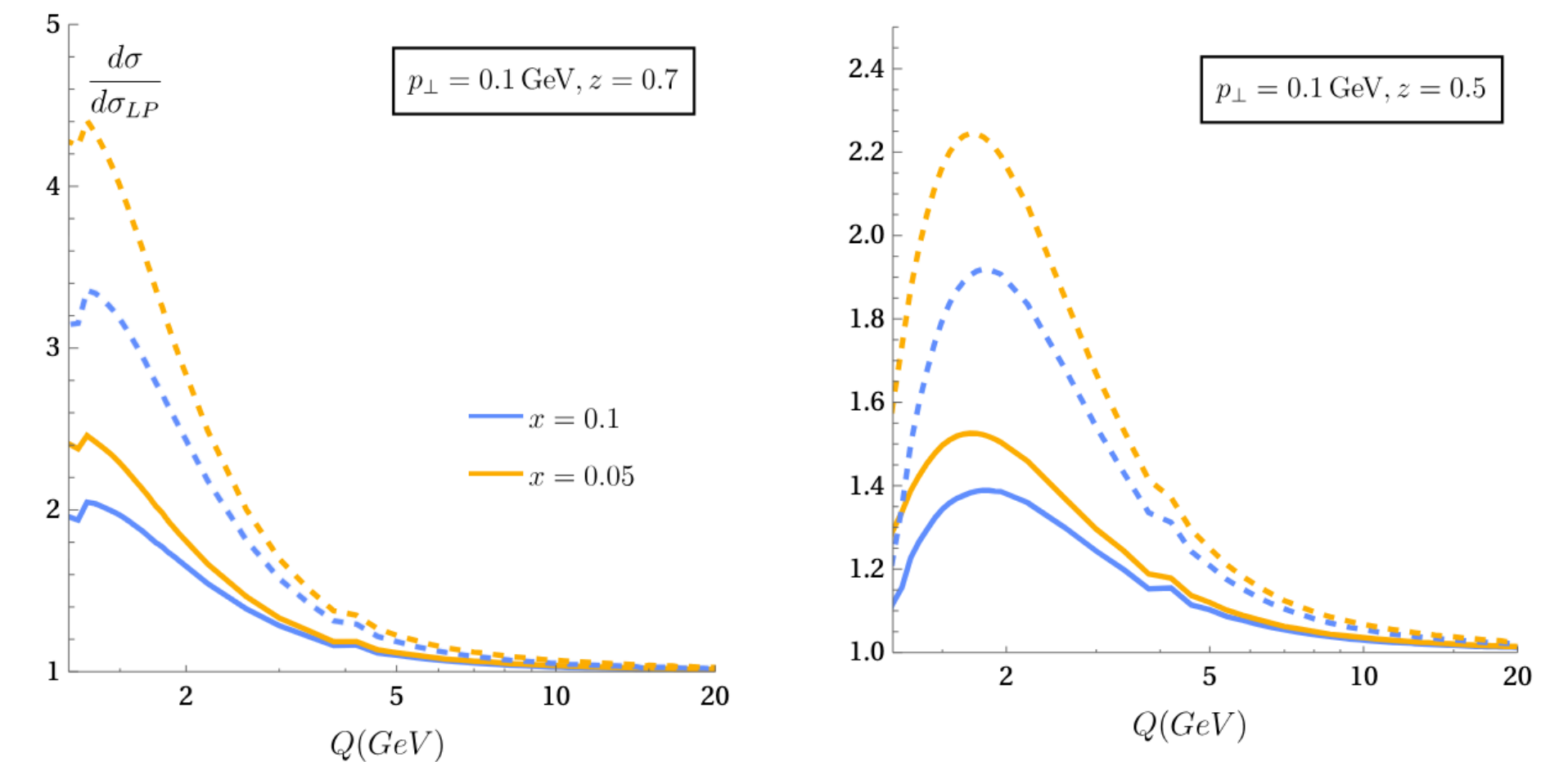
$$F_L = \frac{1}{2z} \int d^2 \vec{P}_{hT} F_{UU,L}^{(FO)}$$



Sub-leading contributions in W term

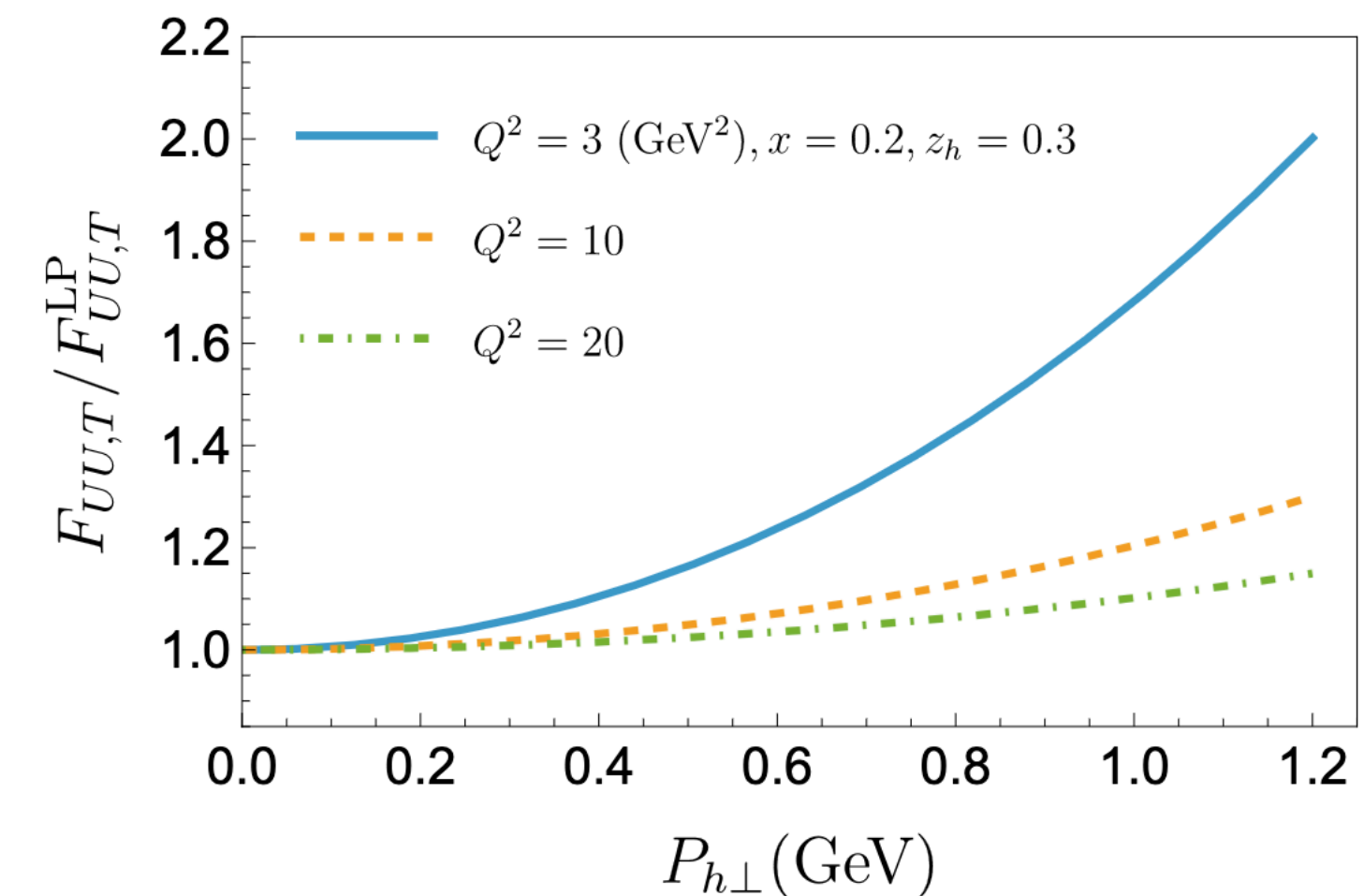
Kinematic power corrections for TMD factorization theorem of semi-inclusive deep-inelastic scattering

Sara Piloñeta^a and Alexey Vladimirov^a



Next-to-next-to-leading power corrections to unpolarized Semi-Inclusive Deep Inelastic Scattering

Ian Balitsky^{a,c} Alexei Prokudin^{b,c}



Probe for sub-leading physics

Truncated SIDIS

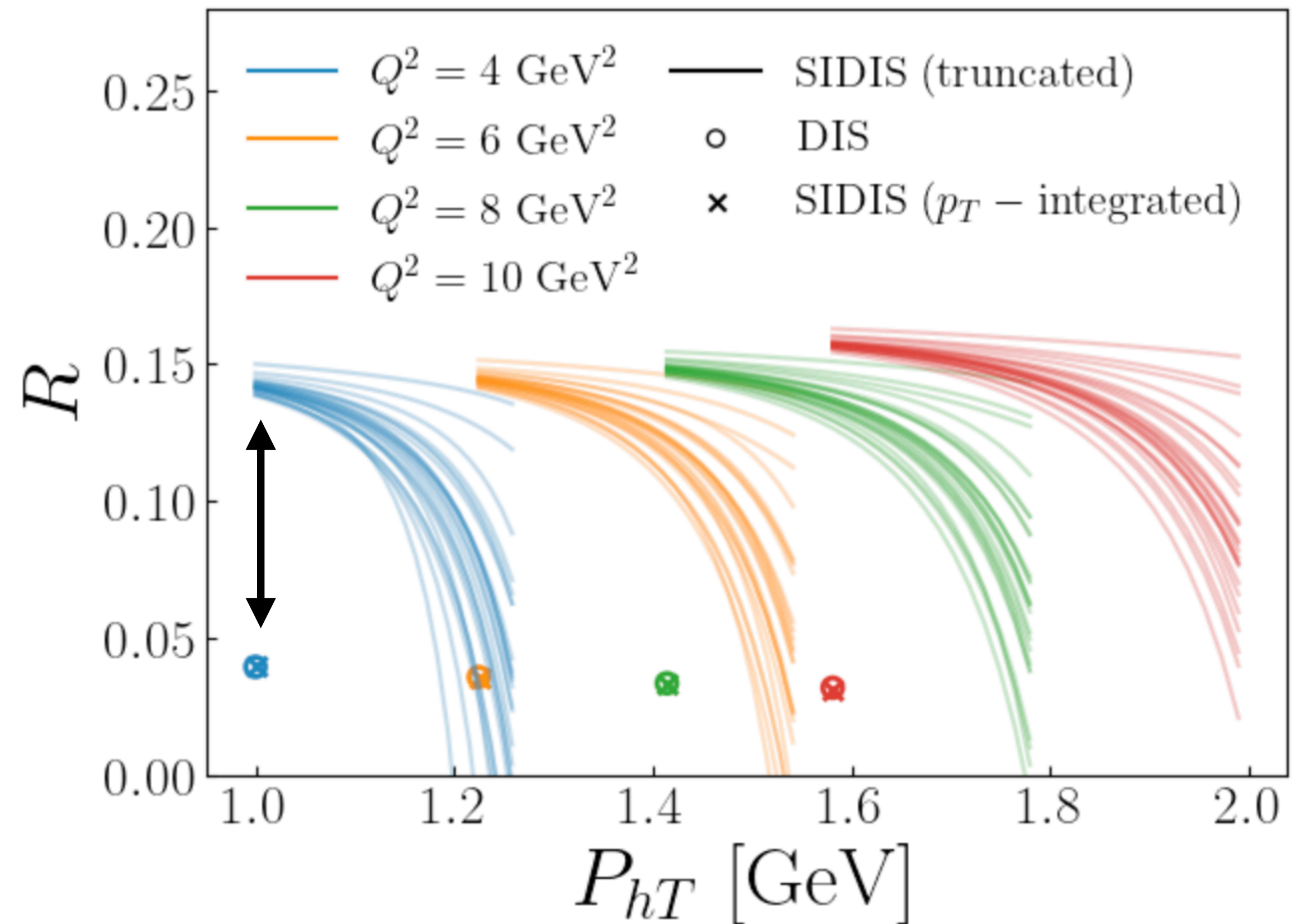
$$R = \int_{P_{hT}} d^2 \vec{P}'_{hT} F_{UU,L}^{(\text{FO})} / \int_{P_{hT}} d^2 \vec{P}'_{hT} F_{UU,T}^{(\text{FO})}$$

Collinear SIDIS/DIS

$$R = F_L / F_T$$

W-term is important for R_{SIDIS} !

What role does sub-leading $F_{UU,L}^{(W)}$ play?



* Upcoming/ongoing JLab experiments: E12-06-104, E12-09-017, E12-09-002

Standard $W + Y$

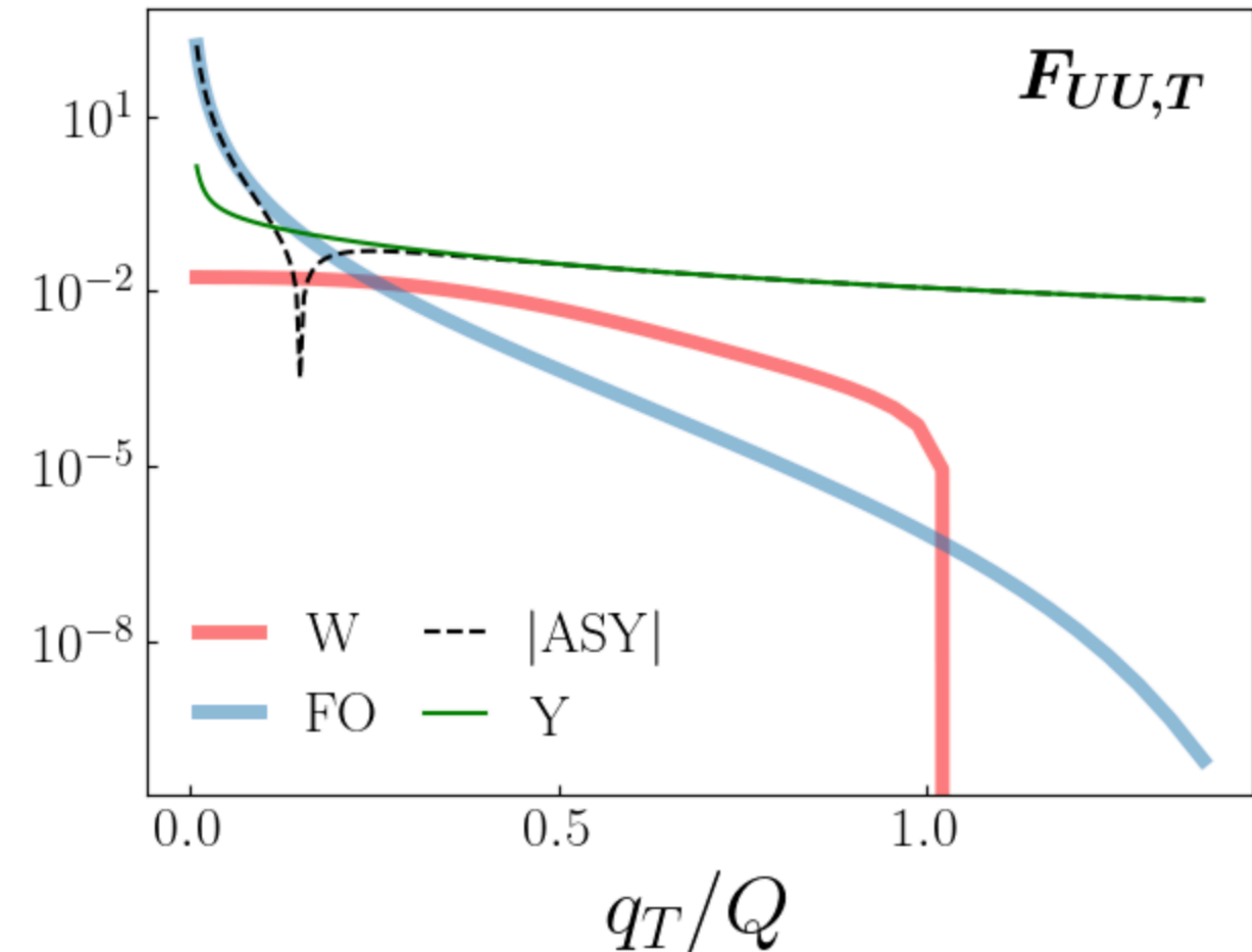
HSO demonstrated issues with “*standard*” $W+Y$

→ theoretical matching from power counting ...

→ in practice: (1) order of magnitude differences, (2) no common “overlap”

Matches and mismatches in the descriptions of semi-inclusive processes at low and high transverse momentum

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Resolution to the problem of consistent large transverse momentum in TMDs

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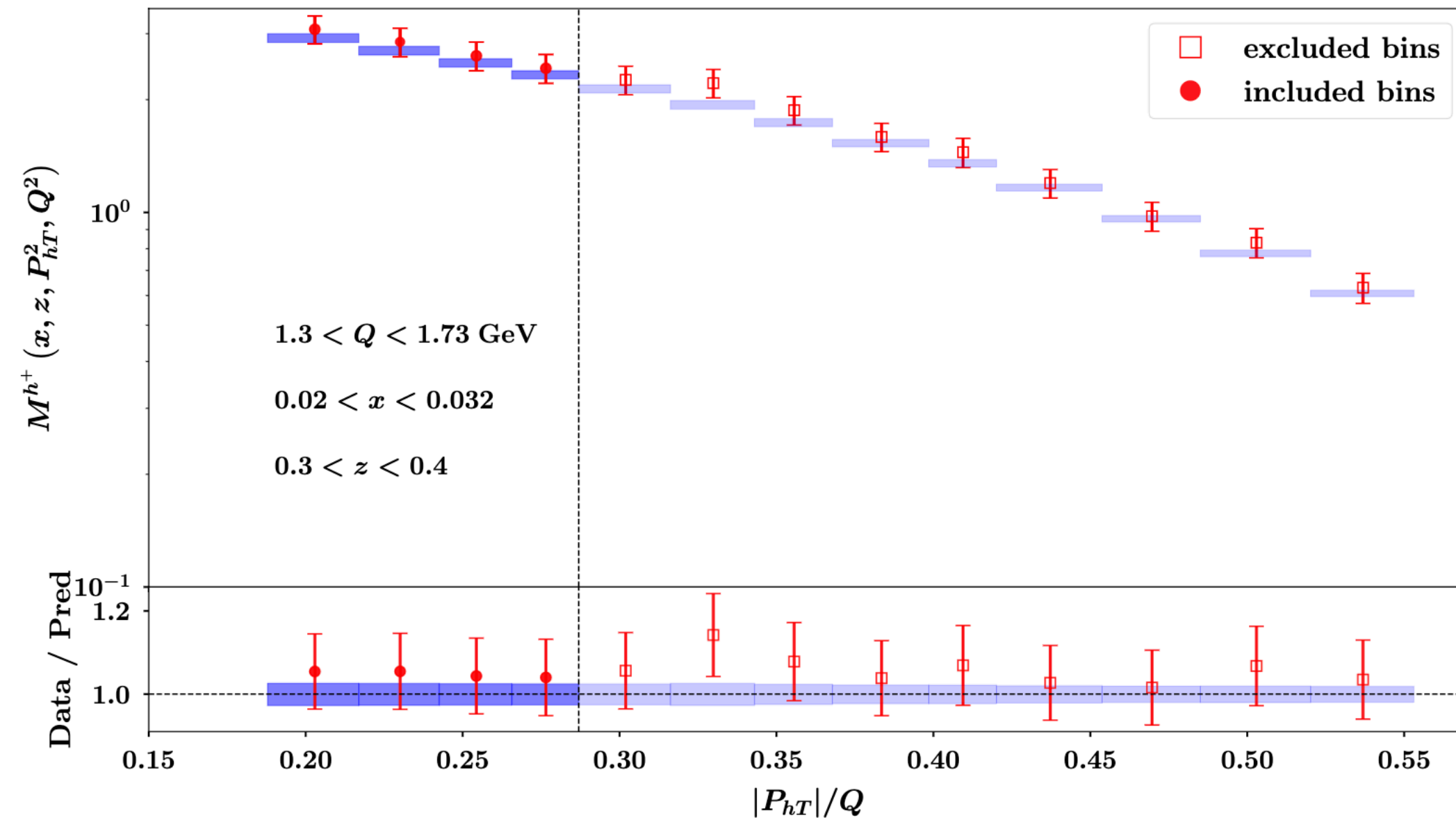
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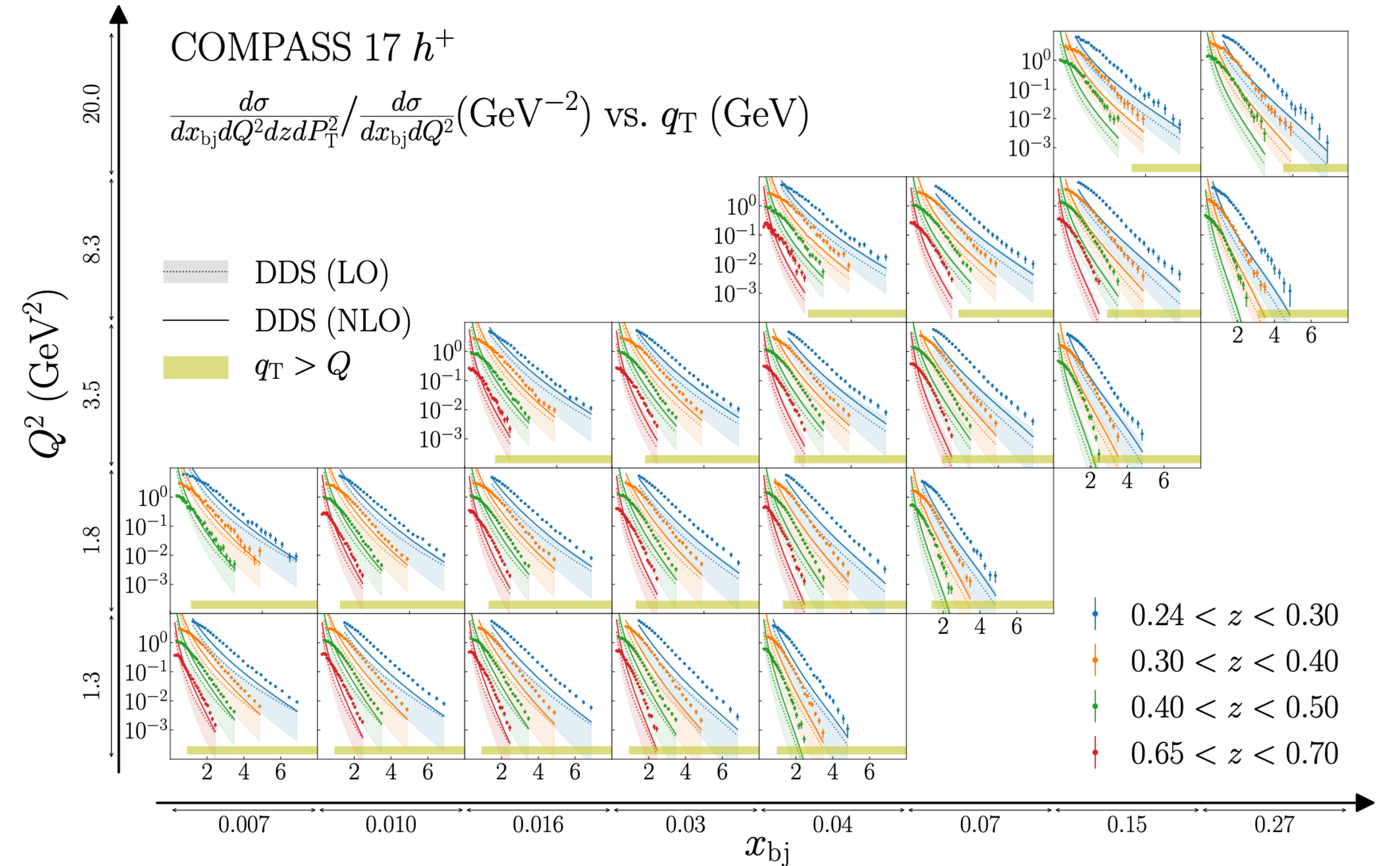
Interplay between regions

Bacchetta et. al., arXiv:2206.07598v2 [hep-ph] (2022)



W term describes “large” P_{hT} data surprisingly well

Gonzalez-Hernandez et. al., arXiv:1808.04396v2 [hep-ph] (2018)



FO theory does not describe “large” P_{hT} data so well

Building our model

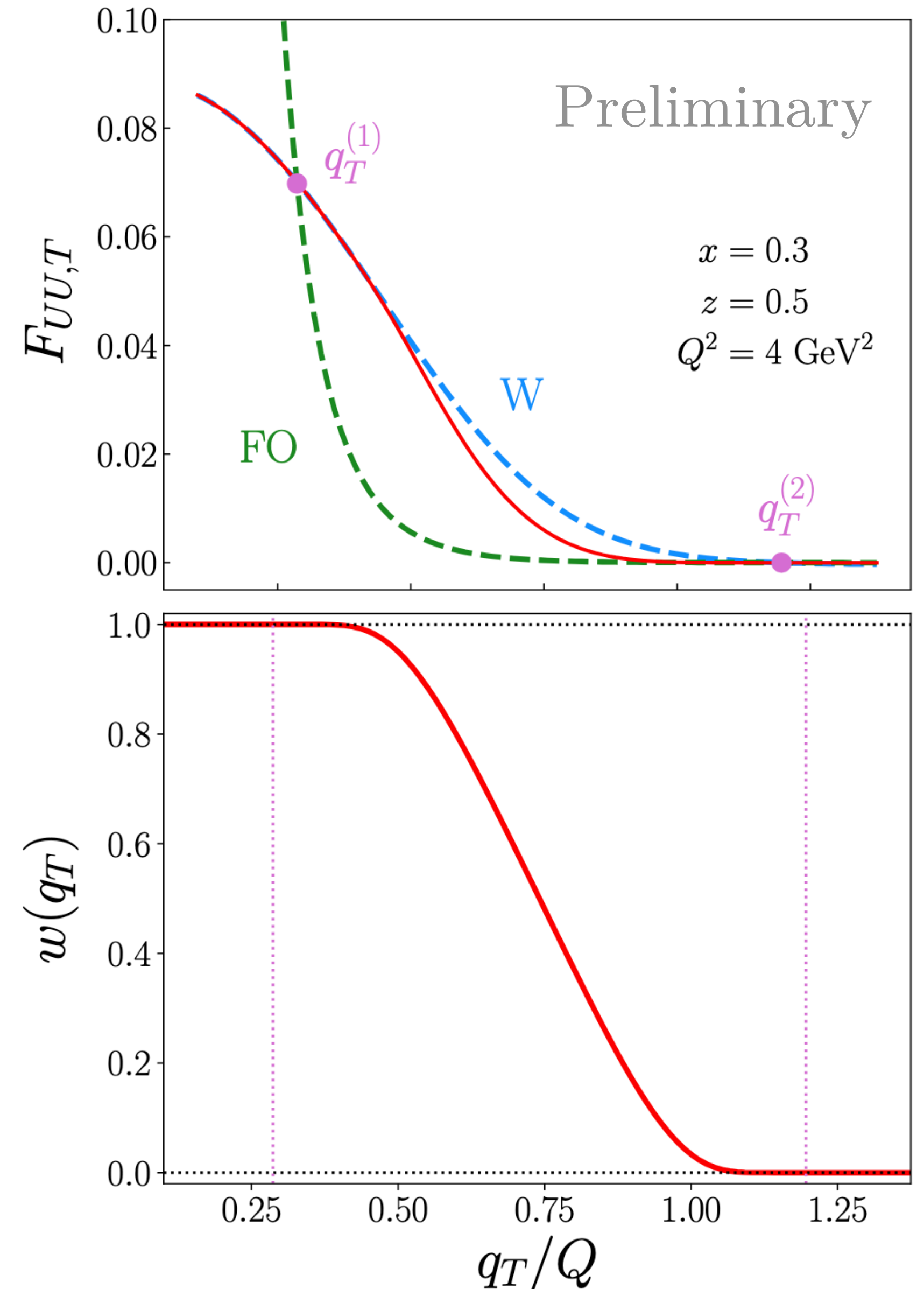
Since $W + Y$ not feasible phenomenologically yet ...

→ approximate ASY by interpolating W & FO

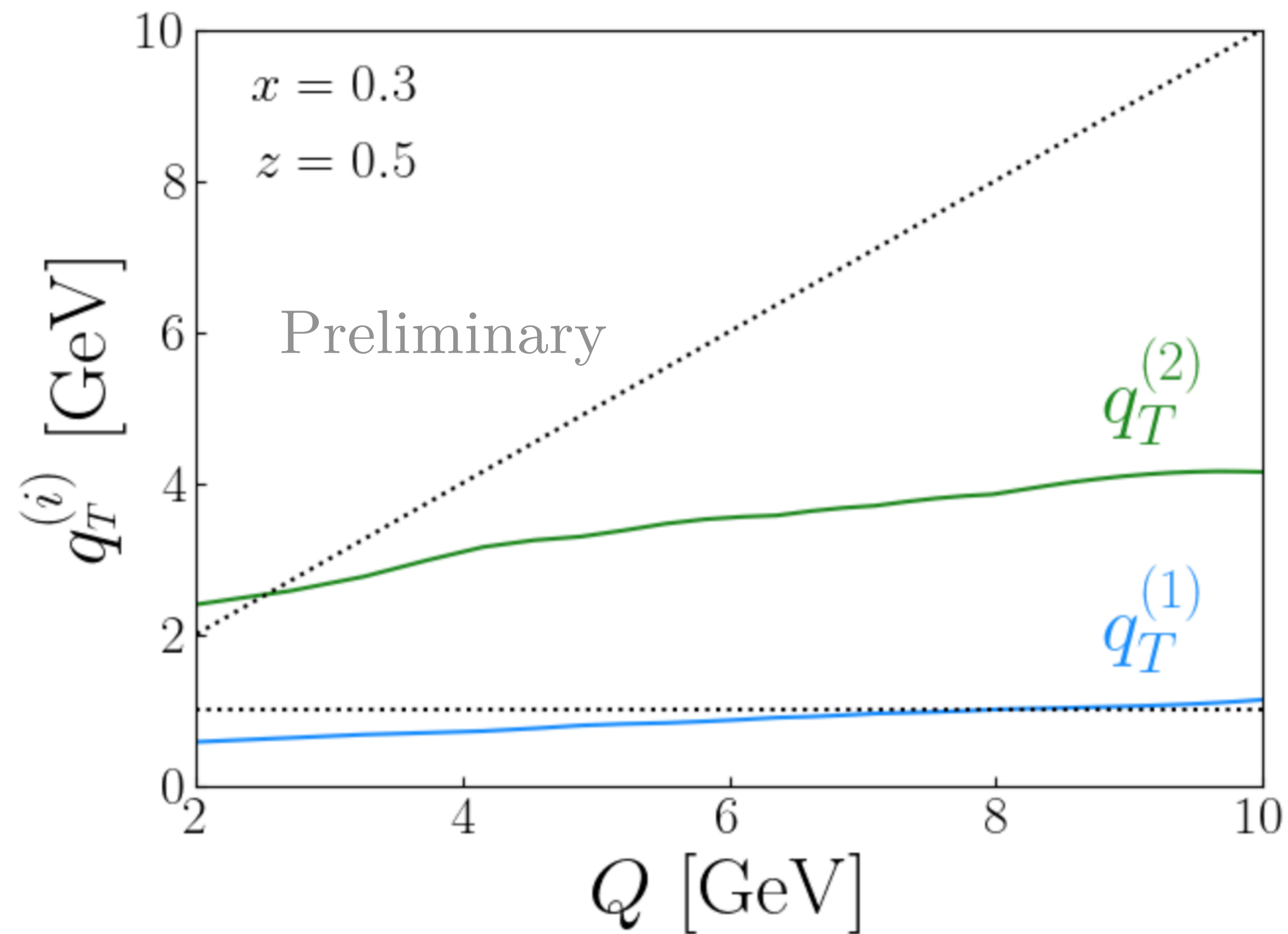
→ use intersection points between W & FO to separate scales

→ ensure analyticity point-by-point

$$F = w(q_T) W + [1 - w(q_T)] FO$$
$$w(q_T) = \begin{cases} 1 & q_T < q_T^{(1)} \\ f_{\text{interp}}(q_T^{(c)}, \sigma) & q_T^{(1)} < q_T < q_T^{(2)} \\ 0 & q_T > q_T^{(2)} \end{cases}$$



Behavior of the intersections



Regions of validity

→ W: $M \sim q_T \ll Q$

→ FO: $M \ll q_T \sim Q$

Intersections roughly follow this trend!

→ weak slope for W term near M

→ stronger slope for FO but < 1

→ intriguing linear behavior ...

Caution: region separation difficult when $M \sim Q$

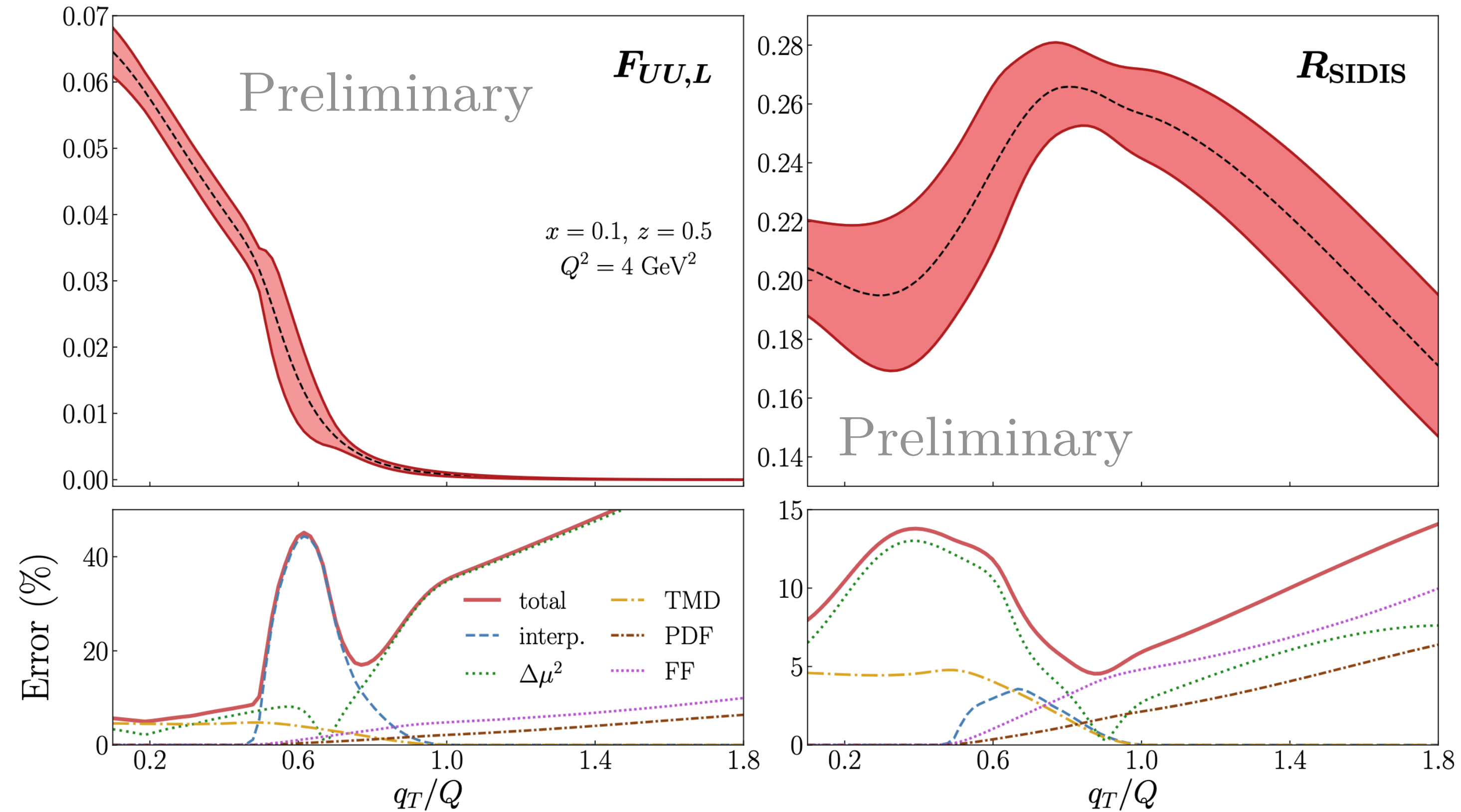
Uncertainty quantification

Use same interpolation for $F_{UU,T/L}$

$$\rightarrow \omega_L = \frac{k_T^2}{k_T^2 + Q^2/4} \rightarrow \frac{k_T^2}{4Q^2} \text{ when } k_T \sim 0$$

$\rightarrow$ 4 types of uncertainty:

- (1) interpolation between W & FO
- (2) Scale variation
- (3) TMD replicas
- (4) Collinear hessian sets



Pheno impacts/opportunities

R_{SIDIS} predictions at Compass kinematics

→ full error budget

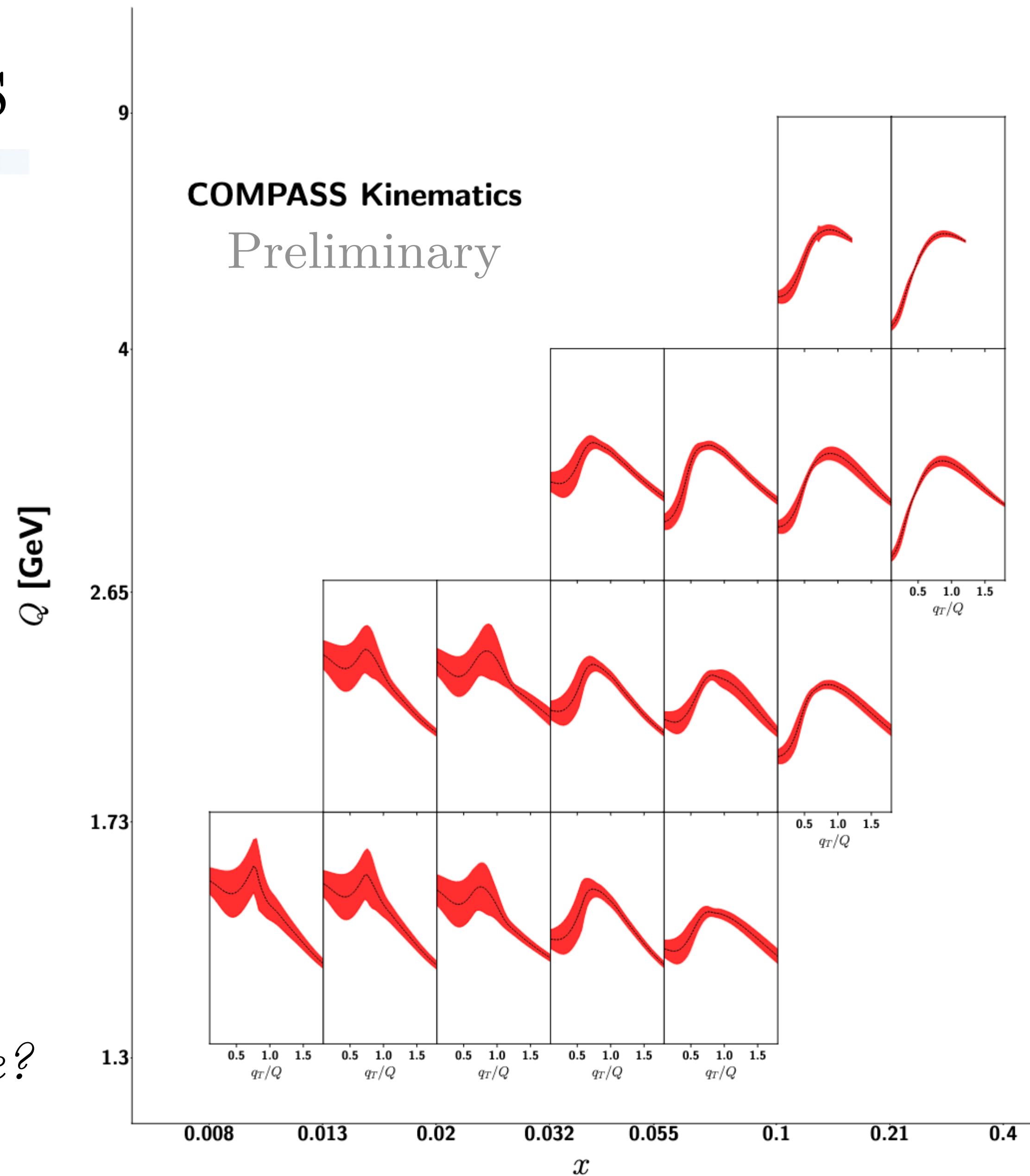
→ trends with increasing Q

(1) Magnitude of $R_{\text{SIDIS}}(q_T \rightarrow 0)$ decreases

(2) $R_{\text{SIDIS}}(q_T \rightarrow 0) \sim (q_T/Q)^2$

(3) Uncertainty decreases

How do theory & experimental uncertainties compare?



Looking forward

SIDIS offers a rich physics landscape

→ indispensable probe of transverse proton structure

→ lots of challenges in formal/
phenomenological description of q_T spectrum

Progress in sub-leading TMD physics:

→ JLab experiments: E12-06-104, E12-09-017, E12-09-002

→ theory: Balitsky & Prokudin, Piloñeta & Vladimirov

This work:

→ pheno estimates with uncertainties of $F_{UU,L}$ and R_{SIDIS} considering interplay of W & FO

→ Future — impacts on multiplicities/collinear SIDIS